JOB PORTAL MANAGEMENT SYSTEM

School of Computing

Bachelor of Technology
in
Information Technology

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BONAFIDE CERTIFICATE

It is certified that the work contained in the Full Stack Application Development report titled Job Portal Management System by Ruhina Naaz K (VTU26453) has been carried out in Winter semester of Academic year 2025-2026(WS2526).

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DECLARATION

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ABSTRACT

The Job Portal Management System is a modern web-based application designed to connect job seekers and employers through a single digital platform. It simplifies the recruitment process by allowing job seekers to create profiles, upload resumes, search for suitable job opportunities, and apply online with ease. At the same time, employers can register their companies, post job vacancies, review applications, and shortlist qualified candidates efficiently. The system provides a user-friendly inter face, secure data management, and quick access to employment opportunities. By automating traditional hiring methods, the project reduces time, effort, and paper work for both applicants and recruiters. This platform serves as an effective solution for improving commun ication between employers and job seekers while making the recruit process faster, transparent, and more organized. The technical backbone of the system is built upon the Spring Boot framework, utilizing a Model-View-Controller (MVC) architecture to ensure a clear separation of concerns and ease of maintenance. Data persistence is managed through MySQL, with Hibernate/JPA facilitating efficient object-relational mapping.

Keywords : Job Portal, Recruitment System, Job Seekers, Employers, Online and Application Resume Upload, Vacancy Posting.

LIST OF FIGURES

LIST OF TABLES

LIST OF ACRONYMS AND ABBREVIATIONS

API	Application Programming Interface
MVC	Model View Controller
JPA	Java Persistence API
ORM	Object Relational Mapping
SQL	Structured Query Language
OTP	One Time Password
REST	Representational State Transfer
SMTP	Simple Mail Transfer Protocol
CRUD	Create Read Update Delete
UI	User Interface

TABLE OF CONTENTS

Chapter 1

INTRODUCTION

1.1 Introduction

The Job Portal Management System is a web-based application to developed simplify and modernize the recruitment process for both job seekers and employers. In today's competitive world, finding the right job and hiring suitable candidates through traditional methods can be time-consuming and inefficient. This system provides a cen tralized online platform where job seekers can create profiles, upload resumes, search for jobs, and apply directly to available opportunities. Similarly, employers can register their organizations, post job vacan cies, review applicant details, and shortlist candidates based on their requirements. The project [1] designed with an easy-to-use interface, secure database management, and efficient communication features [1] ensure a smooth hiring experience. By digitizing the recruitment process, the Job Por tal Management System reduces paperwork, saves time, and improves transparency, making it a valuable solution for modern employment and placement needs. This project not only meets the current requirements for digital job seeking but also provides a scalable framework capable of integrating future advancements such as AI-driven candidate matching and automated screening processes.

1.2 Aim of the Project

The main aim of the Job Portal Management System is to develop an efficient and user-friendly online platform that connects job seeker with employers in a simple and organized manner. The project is designed to streamline the recruitment pro cess by enabling candidates [2] create profiles, upload resumes, search for jobs, and apply online, while allowing employers to post vacancies, manage applications, and shortlist suitable candidates.

It aims to reduce the time, cost, and manual effort involved in these traditional hire methods, improve between recruiters and applicants, and a secure and transparent system for employment opportunities.

1.3 Scope of the Project

The scope of the Job Portal Management System is broad, as it provides a complete digital platform for recruitment and job search activities. Job seekers can create pro files, upload resumes, search jobs based on skills, location, and category, and apply online with ease. Employers can register their companies, post vacancies, manage applications, and shortlist suitable candidates efficiently. The system also includes administrative features such as user management, These monitoring job postings, approval processes, and maintaining database security.

It can be used by colleges, companies, consultancies, and recruitment agencies [2] simplify hiring operations. Furthermore, the project has strong future scope with enhancements such as The AI-based job recommendations, interview scheduling, email and OTP notific, the chatbot assistance, analytics dashboards, mobile applicat support, and cloud deployment, making [2] a scalable and modern solution for the recruitment industry.

1.4 Framework

The Job Portal Management System is developed using modern web technologies and frameworks to ensure high performance, scalability, and user-friendly function ality. The backend framework used is Spring Boot, which simplifies development through built-in configurations, RESTful services, and secure data handling. For the frontend, HTML5, CSS3, Bootstrap, and Thymeleaf are used to create a responsive and attractive user interface across different devices. MySQL [2] used as the database management system to securely store user details, job postings, applications, and other records. Additional tools such as Hibernate/JPA are used for database connec tivity and object-relational mapping. This combination of frameworks provides a to bust archtec, faster development process, easy maintenance, reliability, and flexibility for [5] Job Portal Management System.

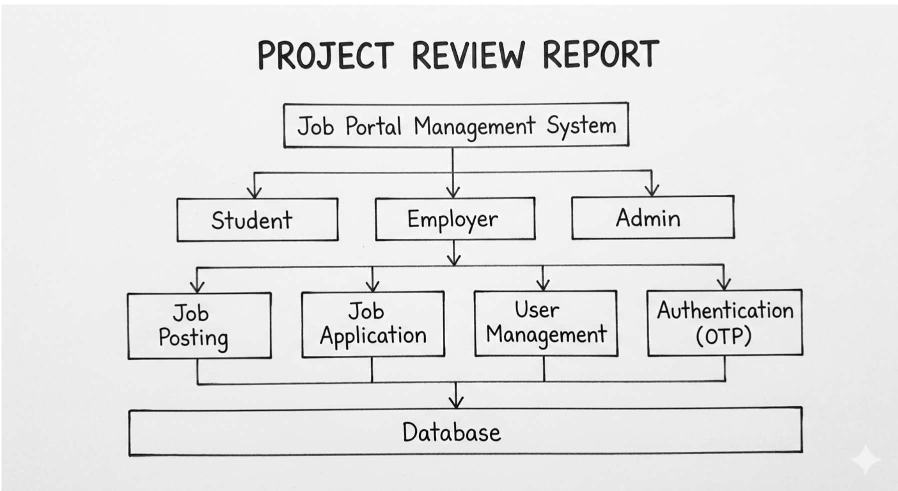

Figure 1.1: System Framework Architecture

Chapter 2

SURVEY OF SIMILAR APPLICATION IN MARKET

In the current digital era, many online job portal applications are available in the market that connect employers with job seekers and efficiently. These platforms provide services such as job searching, resume uploading, company reviews, in terview preparation, and online recruitment management. Popular job portals like LinkedIn, Indeed, Naukri, Glassdoor, Monster, and Shine have transformed the for ditional hiring process into a faster and more transparent digital system. These applications are widely used by companies, startups, consultancies, and educational institutions [6] recruit suitable candidates for various roles. LinkedIn is highly popular for professional networking and the career opportunities, while Indeed is known for its large number of job listings and global reach. Naukri is one of the leading portals in India, especially for freshers and experienced professionals. Glassdoor com bines job listings with company reviews and salary insights, helping users make informed decisions before applying. Monster and Shine also provide resume services, career guidance, and employer branding tools.

Although these applications are successful, many of them still face limitations such as fake job postings, complicated application from processes, excessive will be advertisements, lack of personalized and delayed recruiter responses. Some platforms require premium subscriptions for better visibility, while smaller companies may find [5] pricing expensive.

Popular Similar Applications

- **LinkedIn** – As the world's largest professional networking platform, LinkedIn goes beyond simple job listings. It allows users to build a digital professional brand, engage with industry-specific content, and connect directly with recruiters through "InMail." Its "Easy Apply" feature and AI-driven job recommendations make it a primary tool for modern career growth.

- **Indeed** – Known as a comprehensive job search engine, Indeed aggregates millions of listings from thousands of websites, including company career pages and job boards. It provides robust tools for salary comparison, company to views, and a simplified resume-builder that helps candidates apply to multiple positions with a single click.

- **Naukri** – A dominant force in the Indian recruitment market, Naukri offers a vast database of openings across diverse sectors like IT, Banking, and Manu facturing. It provides specialized services such as "Naukri FastForward" for resume writing and profile enhancement, ensuring candidates stand out to premium recruiters.

- **Glassdoor** – This platform revolutionized the job search by introducing transparency. It provides crowdsourced insights into company cultures, CEO to proval ratings, and detailed salary data. By allowing candidates to read about real interview experiences, it helps job seekers prepare more effectively for the hiring process.

Chapter 3

FRAMEWORK DESCRIPTION

3.1 Frontend Implementation

The frontend implementation of the Job Portal Management System is developed using HTML5, CSS3, Bootstrap, JavaScript, and Thymeleaf to provide an interac tive, responsive, and user-friendly interface. HTML5 is used to structure [8] web pages, while CSS3 and Bootstrap are used for styling, layout design, and mobile responsiveness across different devices. JavaScript is to add dynamic features such as form validation, search filtering, alerts, theme switching, and smooth user inter actions. Thymeleaf is integrated with Spring Boot to render dynamic data from the backend into [8] frontend pages efficiently. The frontend includes modules such as home page, login and registration pages, The OTP verification, job listings, admin dashboard, employer dashboard, and job seeker dashboard and attractive design.

A critical component of the frontend is **Thymeleaf**, the server-side Java template engine. Unlike static HTML, Thymeleaf allows for the dynamic rendering of data attributes. It also handles conditional rendering, such as displaying the ”Post a Job” button only for users logged in with an ’Employer’ role, while hiding it from regular ’Job Seekers.

3.1.1 Environment Setup

The development environment for the Job Portal Management is configured with the necessary software and tools to ensure smooth design, coding, testing, and to ployment of the application. The project is developed using Java as the program language with Spring Boot as the backend framework for creating secure and scalable web services. Eclipse IDE or IntelliJ IDEA is used as the integrated development environment for coding and project management.

The frontend, HTML5, CSS3, Bootstrap, JavaScript, and Thymeleaf are used to build a responsive and interactive user interface. The database used for the project is MySQL, which stores user accounts, company details, job postings, resumes, and application records securely. MySQL Workbench can be used for database design and administration.

3.1.2 Structure of the Project

The structure of the Job Portal Management System is organized in a modular manner to ensure easy development, maintenance, and scalability. The project follows a layered architecture consisting of presentation layer, business logic layer, and database layer. The pre sentation layer includes the user interface developed using HTML5, CSS3, Bootstrap, JavaScript, and Thymeleaf, where job seekers, and employers, and administrators interact with the sys tem through web pages. The business logic layer is developed using Java and Spring Boot, which handles all core functionalities such as user registration, login authentication, job posting, to sure upload, application process, candidate shortlisting, notifications. This layer also manages security, and communication between [8] frontend and database.

3.1.3 Execution Procedure

The execution procedure of the Job Portal Management begins with setting up the required development environment, including Java JDK, Spring Boot, MySQL, Maven, and an IDE such as Eclipse or IntelliJ IDEA. First, the project source code is imported into the IDE, and all required dependencies are downloaded autome using a Maven.

Next, the MySQL database is created, and the necessary and tables for users, to ployers, job postings, resumes, and applications are configured. Database connection details such as username, pass word, and URL are updated in the application config uration file. After all completing the setup, the Spring Boot applica tion is executed from the main class or by using the Maven run command.

Once the server starts successfully, the system can be accessed through a web browser using the local host address. Users can then register as job seekers or employers, log in to surely, and access their respective dashboards. Job seekers can search and apply for jobs, while employers can post vacancies and manage applications. The administrator can monitor the overall system, manage users, and maintain data integrity.

3.2 Backend Implementation

3.2.1 Environment Setup

The environment setup for the Job Portal Management formly it includes in stalling and configuring all required software, tools, and technologies needed for development and execution. The form project uses Java Development Kit (JDK 17 or above) as the pro gramming platform and Spring Boot as the backend framework for developing secure and scalable web applications.

An Integrated Development Environment such as Eclipse IDE or IntelliJ IDEA is used for coding, debugging, and project management. For the frontend develop ment, HTML5, CSS3, Bootstrap, JavaScript, and Thymeleaf are used to design a re sponsive and user-friendly web pages. The database management used is MySQL, which stores information such as user profiles, job posted, employer details, resumes, and applications. MySQL Workbench can be used for creating and managing the database schema.

3.2.2 Structure of the Project

The presentation layer contains the user interface developed using HTML5, CSS3, Bootstrap, JavaScript, and Thymeleaf. It includes web pages for user registration, login, job search, job posting, application forms, dashboards, and admin manage tent. This layer allows job seekers, employers, and administrators to interact with the system easily. The application and business logic layer is developed using Java and Spring Boot. It includes modules such as controllers, services, reposi tories, and security configure.

Controllers manage user requests, services handle business from operations, repos itories manage database as access, and security modules provide authentication and authorization. Core functions like resume upload, job posting, application tracking, candidate shortlisting, and notifications are handled in this layer. The database layer uses MySQL with Hibernate/JPA for storing and managing records such as the user details, company information, job vacancies, resumes, and application status.

3.2.3 Execution Procedure

The execution procedure of the Job Portal Management System starts with in stalling the required software such as Java JDK, Spring Boot, MySQL, Apache Maven, and an IDE like Eclipse or IntelliJ IDEA. After installation, the project source code is imported into the IDE, and Maven automatically downloads all neces sary need dependencies and libraries required for the application. Next, the MySQL database is created, and the required tables for users, employers, job postings, re sumes, and job applications are other configured. The database connection details such as database name, username, password, and server port are updated in the application configuration file (application.properties or application.yml). Once the database setup is completed, the project is ready for execution.

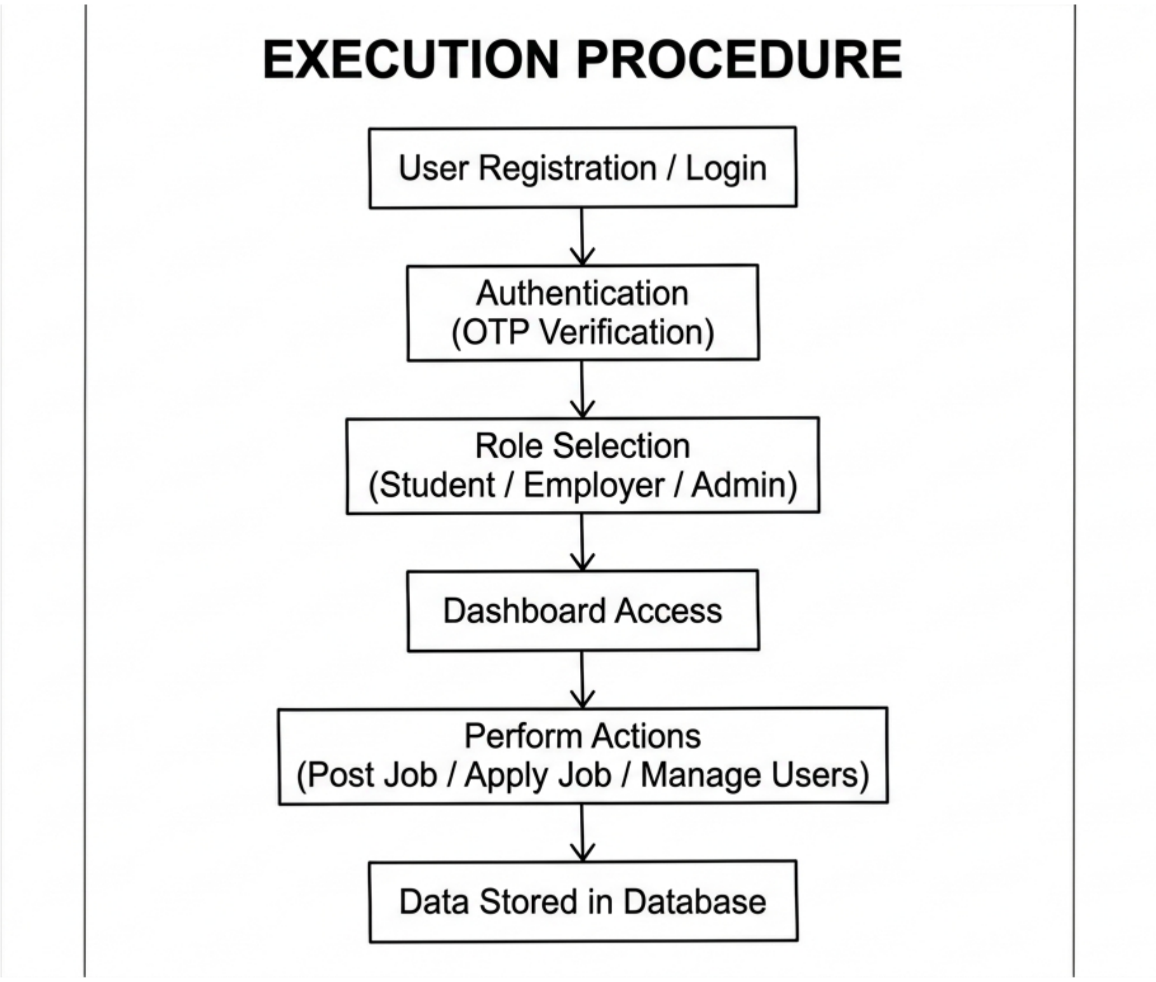

Figure 3.1: Execution Procedure

The execution procedure emphasizes the importance of environment consistency, ensuring that dependency versions are strictly managed to avoid runtime conflicts. By the following this structured deployment cycle, the system undergoes a series of internal validation checks, such as verifying the persistence layer's connectivity and the responsiveness of the RESTful endpoints. This such as cloud integration or microservices migration. Ultimately, this rigorous execution ensures that the Job Portal Management System.

3.3 Database Implementation

3.3.1 Database Configuration

The database configuration of the Job Portal Management System is designed to ensure secure storage, fast retrieval, and efficient at management of application data. The project uses MySQL as the are relational database management system to store information related to job seekers, employers, job postings, resumes, applications, and administrative records.

A dedicated database is created for the system, and all The required tables are with proper primary keys, for eign keys, and constraints to maintain data integrity. The connection between the application and database is established using Spring Boot configuration files such as application.properties or application.yml.

Important parameters including database URL, username, password, driver class name, and Hibernate settings are specified in the config file. Hibernate/JPA is used for Object Relational Mapping (ORM), which simplifies database operations like insert, update, delete, and retrieval through Java entities and repositories.

3.1.2 Schema Structure

The schema structure of the Job Portal Management System is the designed to organize and manage all application data in a relational database efficiently. The project utilizes a MySQL database where interconnected tables are structured with primary keys, foreign keys, and constraints to maintain data integrity.

a) User Management: The primary Users table stores comprehensive details of job seekers, including unique user IDs, The names, encrypted passwords, contact information, and professional skill sets.

b) Employer and Job Profiles: The Employers table manages company related data, which is linked to the Jobs table. This allows for detailed job vacancy tracking, including titles, categories, salary ranges, and experience requirements.

c) Application Tracking: The Applications table serves as a relational between job seekers and vacancies, recording application dates and real-time statuses such as 'Shortlisted' or 'Selected'.

d) Resource Storage: A dedicated Resumes table manages the file for uploaded documents, ensuring that user profiles are correctly associated with their professional curriculum vitae.

e) Administrative Control: The Admin table stores secure login credentials and management details, allowing for high-level oversight of the platform's users and content.

This organized schema ensures efficient data storage, faster query processing, and secure management across the entire platform. By the enforcing relational constraints, these are system minimizes data redundancy and provides a smooth op erational flow for both recruiters and applicants. Furthermore, this modular database design allows for seamless future scalability, such as the addition of advanced analytics or AI-driven recommendation modules.

3.1.3 Tables created

The Job Portal Management System database includes several tables created to manage and store all required information efficiently. Each table is designed for a specific purpose and connected through the relationships to maintain data consistency. The main tables created in the system are Users, Employers, Jobs, Applications, Resumes, and Admin.

The Users table stores details of job seekers such as user ID, full name, email address, password, phone number, skills, education, and experience, and profile in formation. The Employers table contains company details such as employer ID, company name, email, pass word, contact number, address, and company description.

Chapter 4

METHODOLOGIES

4.1 Project Architecture

The architecture of the Job Portal Management System follows a multi-layered structure to ensure efficient performance, scalability, and maintainability. The system consists of three main layers: presentation layer, business logic layer, and database layer. The presentation layer is developed using Techstack HTML5, CSS3, Boot strap, JavaScript, and Thymeleaf, providing an interactive interface [14] job seekers, em ployers, and administrators.

The business logic layer is built using Java and Spring Boot, which manages core functionalities such as user registration, login authentication, job posting, to sure upload, application processing, candidate shortlisting, and notifications. The database layer uses MySQL with Hibernate/JPA to store and manage data related to users, companies, job vacancies, resumes, and applications securely. Controllers handle user requests, services process application logic, repositories interact with [13] database, and security modules ensure authorized access. This architecture in ables smooth communication between all modules and supports future enhancements such as AI based recommend, chatbot integration, email alerts, and analytics in the dashboards.

4.2 DataFlow Diagram

The Data Flow Diagram (DFD) of the Job Portal Management and System the presents the movement of data between users, processes, and the database within the system. The main external entities involved are Job Seekers, Employers, and in min. Job seekers provide input such as registration details, login credentials, resume uploads, and job, while employers submit company information, job postings, and candidate selection details.

The system processes this information through mod ules such as user authentication, profile manage, and job management, application processing, and admin control. All processed data is stored in the MySQL database, which contains tables [11] users, employers, jobs, resumes, and applications. The admin monitors user ac tivities, manages job listings, and maintains system security. The DFD clearly shows how data enters the system, gets processed, stored, and retrieved to provide efficient interaction between all users and modules of [13] Job Portal Management System.

4.3 Module 1: User Registration and Authentication

This module is responsible [11] allowing job seekers, employers, and adminis trators to create accounts and securely access the Job Portal Management System. New users can register by providing details such as name, email, password, contact number, and role selection. After successful registration, users can log in using valid credentials. The system verifies login details, encrypts passwords, and grants access to the respective dashboard based on user role. It also includes features such as forgot password, profile update, logout, and session manage to ensure secure and smooth user access.

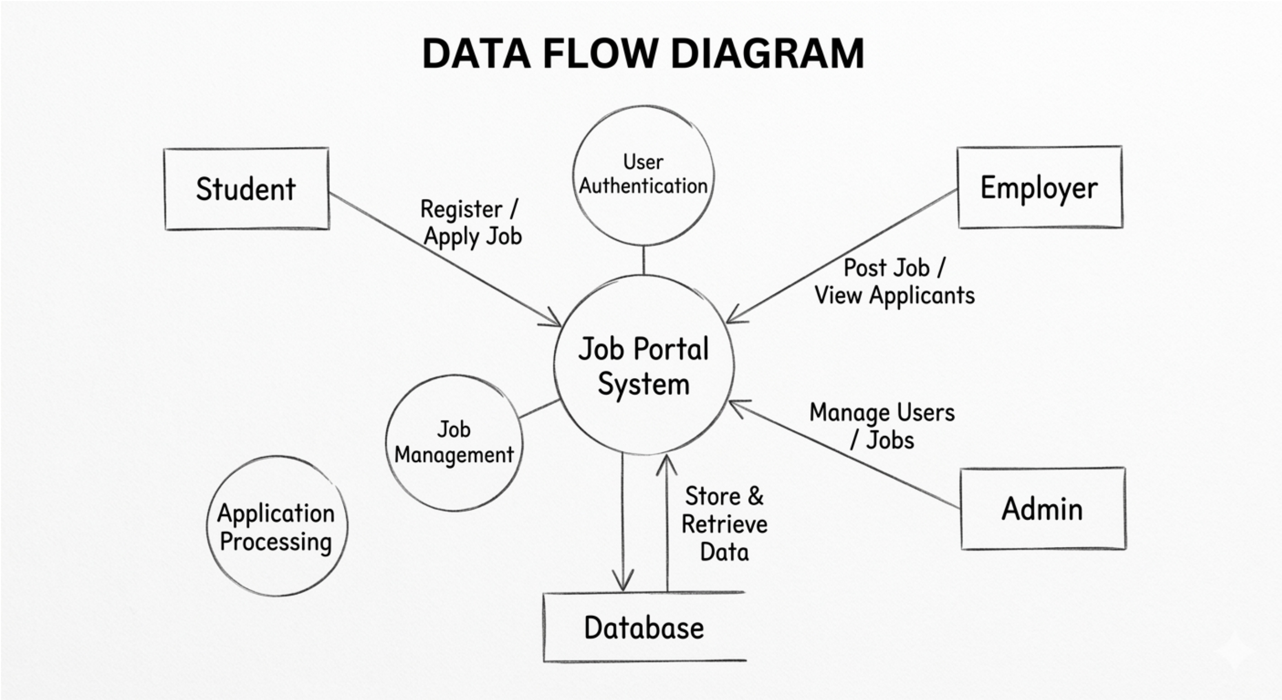

Figure 4.1: Dataflow Diagram

4.4 Module 2: Job Seeker Management

This module is designed for job seekers to manage their profiles and access employment opportunities through the Job Portal Management System. Users can create and update personal profiles by adding by the educational qualifications, skills, experience, certifications, and contact details. Job seekers can upload resumes, search for jobs based on category, location, or company, and apply [14] suitable vacancies on line. They can also track application status such as applied, shortlisted, rejected, or selected. This module helps candidates efficiently manage their job search process and improve their chances of employment.

4.5 Module 3: Employer Management

This module is designed for employers and companies to manage recruitment activities efficiently through the Job Portal Management System. Employers can create company profiles by adding organize details, contact information, and company descriptions. They can post new job such as job title, skills required, salary, experience, and llocation. Employers are able to view received applications, review candidate to sures, shortlist suitable applicants, schedule interviews, and update hiring status. This module helps organizations streamline the hiring process and find qualified candidates quickly and effectively.

Advantages of this Project

The Job Portal Management System offers several advantages for job seekers, employers, and administrators by simplifying [9] most recruitment process through a digital platform. It reduces [14] time and effort required in traditional hiring methods by enabling online registration, job posting, and instant applications. Job seekers can they easily search for jobs, upload resumes, and track application status from any where, while employers can quickly post vacancies, review applications, and short list candidates efficiently.

a) Easy Job Search: The system allows job seekers to search and apply for jobs quickly based on skills, category, company, and location, saving time and effort.

b) Efficient Recruitment: Employers can post vacancies, review applications, shortlist candidates, and manage hiring activities in an organized manner.

c) Secure Data Management: The project stores user profiles, resumes, and job details securely with proper authentication and the database management.

d) Real-time Application Tracking: One of the standout features of this system is the ability for job seekers to monitor their application status in real-time. Instead of waiting indefinitely for an email, candidates receive immediate updates when an employer views, shortlists, or updates their status. This transparency builds trust and improves the overall user engagement of the platform.

Disadvantages

The Job Portal Management System also has some disadvantages despite its ben efits. The system depends completely on internet and connectivity, so users may face difficulties in areas with poor network access. Initial development, hosting, and maintenance costs may be high [9] small organizations. Regular software updates and security monitoring are required to protect user data from cyber threats. Some job seekers and employers who are not familiar with digital platforms may find the system difficult to use at first. In addition, high and large numbers of applications may make it harder [9] candidates to get noticed quickly.

a) Internet Dependency: These are system depends completely on internet the nectivity; consequently, users may face significant difficulties or service interruptions in areas with poor network access or during server downtime.

b) High Implementation Costs: Initial development, hosting, and professional maintenance costs may be These relatively high for small organizations or startups, required a dedicate budget for infrastructure.

c) Security and Maintenance: Regular software updates and the continuous security monitoring are mandatory to protect sensitive user data and resumes from evolving cyber threats and unauthorized access.

d) Potential Data Privacy Risks: As the platform handles sensitive personal in formation, including contact details and detailed professional histories, it becomes a potential target for data breaches. Despite implementing security protocols, the on-going risk of sophisticated phishing attacks or unauthorized access requires constant vigilance and advanced encryption methods to maintain user trust.

Chapter 5

RESULTS AND DISCUSSIONS

The Job Portal Management System was successfully developed and tested to provide an efficient platform for connecting job seeker and employers. The system performed all major functions such as user registration, secure login, profile management, job posting, resume upload, job search, online application submission, and candidate short listing accurately. The database stored and retrieved user and job in information efficiently, ensuring smooth system performance. During testing, the application showed a user-friendly interface, fast response time, and reliable data handling. Job seekers were able to search and apply [14] jobs easily, while employers could manage vacancies and review applications effectively. The admin module monitored users and maintained system control. The results indicate that the project reduces manual effort, saves time, improves transparency, and sim plify the recruitment process. Overall, the system scalable, and practi cal [11] modern online hiring needs. The results indicate that the project significantly reduces the manual effort typically associated with recruitment. By automating the application tracking process, the system saves time for HR departments by organizing candidate data into a searchable format. For job seekers, the transparency provided by the ”Application Status” feature eliminates the uncertainty of the hiring cycle.

Table 5.1: Comparison of Existing Applications with Proposed Job Portal System

Applications	Advantages
LinkedIn	Professional Networking
Naukri.com	Large Job Listings
Indeed	Easy Job Search
Shine.com	Resume Visibility
Proposed Job Portal System	Fast, Secure & User-Friendly

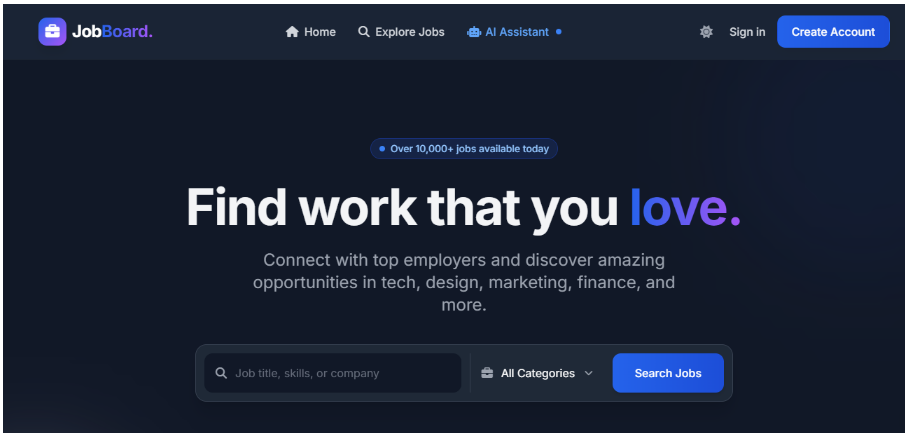

Figure 5.1: Homepage index

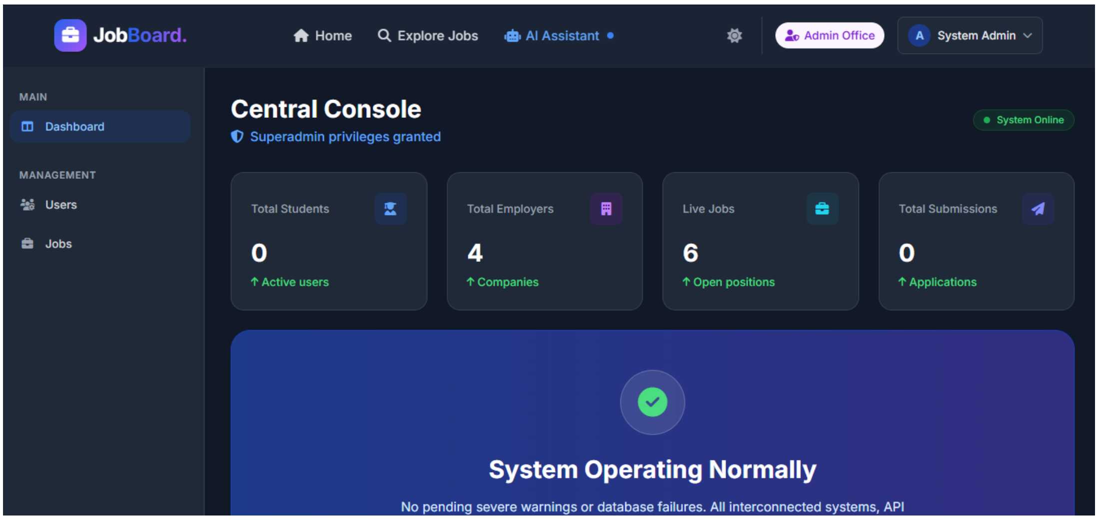

Figure 5.2: Admin Dashboard

The Admin Dashboard provides centralized control for managing users, job postings, and overall system operations. It allows administrators to monitor platform activity and maintain data integrity.

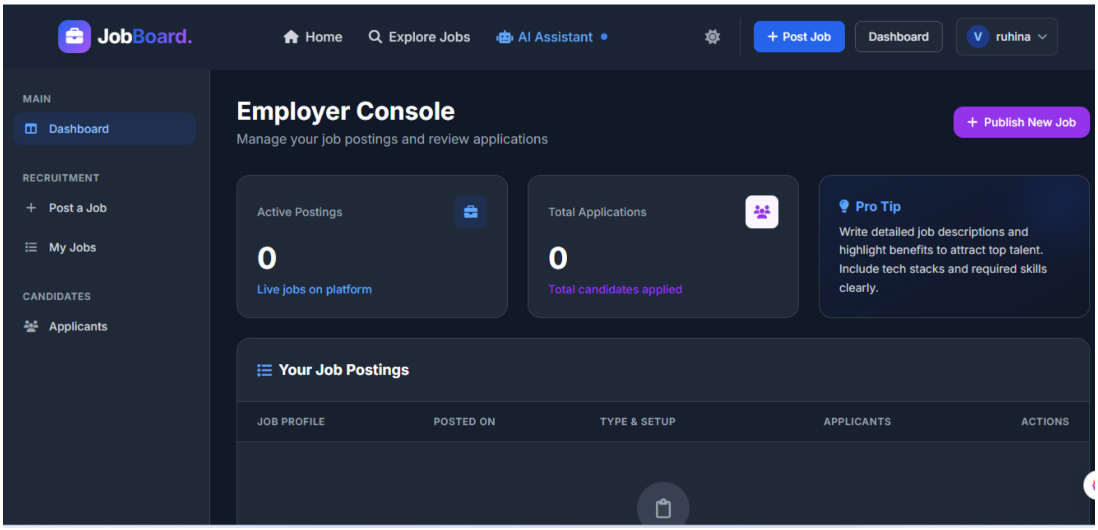

Figure 5.3: Employer Dashboard

The Employer Dashboard enables recruiters to create the job listings, manage the applications, and track candidate progress. It offers an intuitive interface for handling recruitment workflows efficiently.

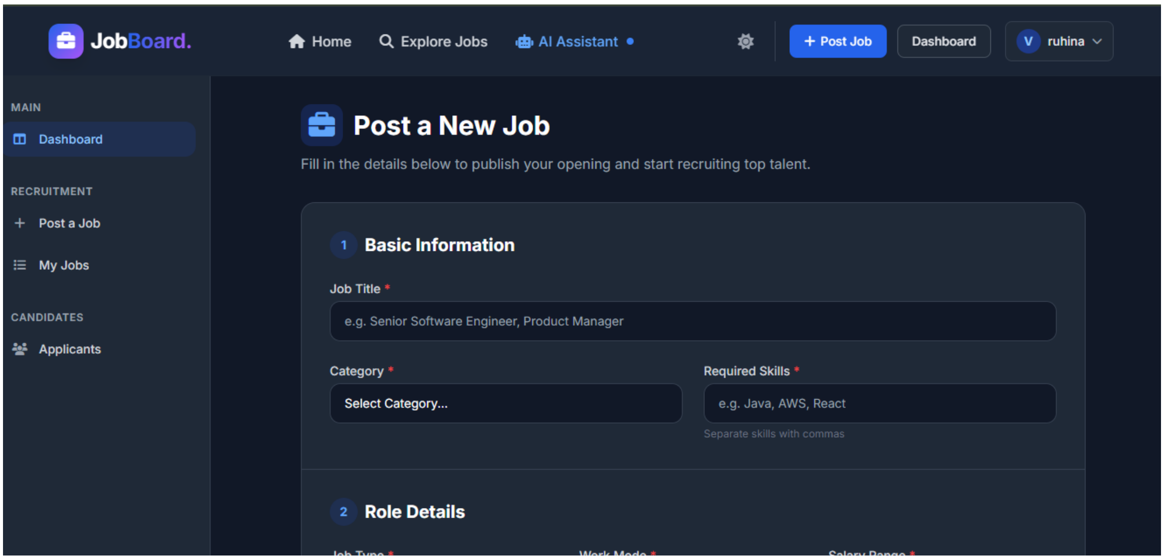

Figure 5.4: Job Posting

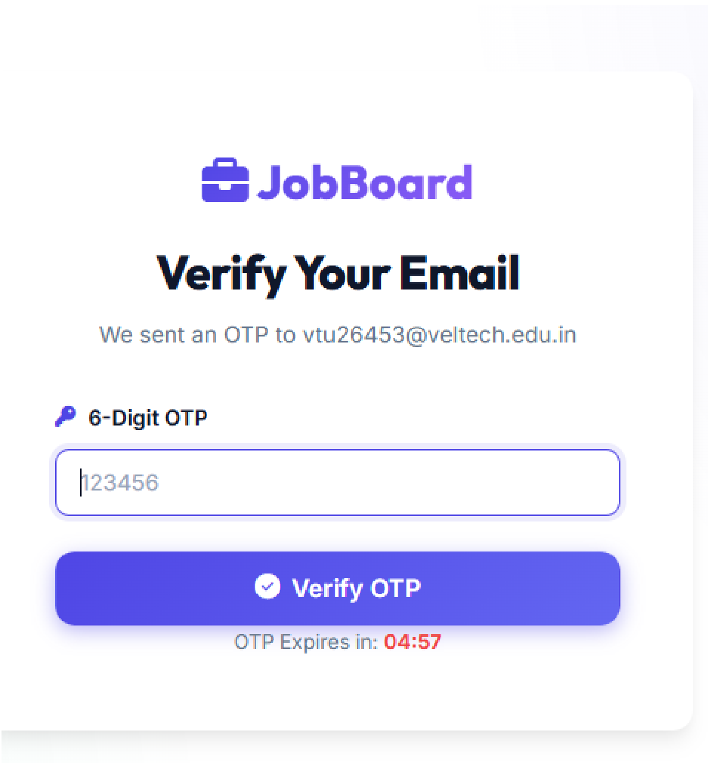

Figure 5.5: OTP Verification

The OTP Verification system enhances security by validating user from identity the during registration and these are login processes it prevents unauthorized.

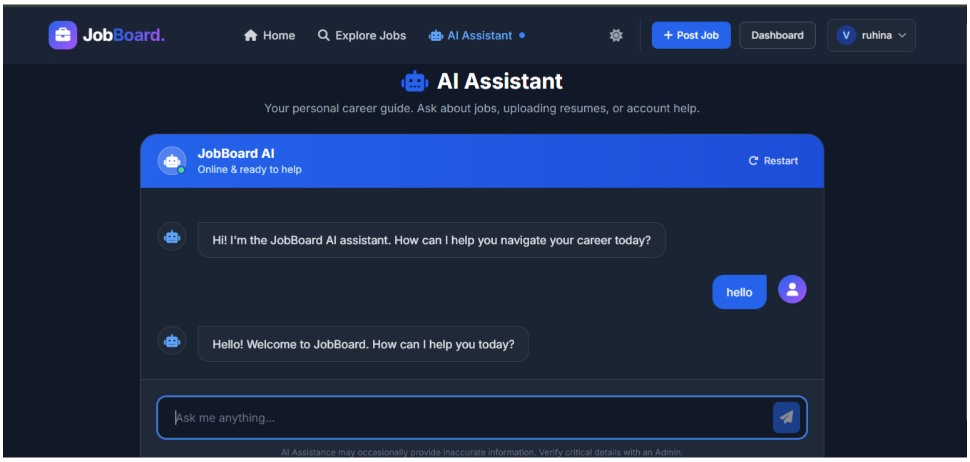

Figure 5.6: Chatbot Assistant

The Chatbot Assistant provides real-time support by answering user queries and guiding them through various platform features. It is the improves user interaction and reduces dependency on manual assistance.

Chapter 6

CONCLUSION AND FUTURE ENHANCEMENTS

6.1 Conclusion

The Job Portal Management System is a successful web-based ideas the solution developed to simplify and modernize the recruitment process [11] job seekers and employers. The system effectively performs there important functions such as user are registration, secure login will profile management, job posting, resume upload, online applications, and candidate shortlisting. It reduces manual effort, saves time, in proves transparency, and provides organized data management through a secure platform.

Overall, the system proved [12] be reliable, scalable, and practical for real-world hiring needs, and it can be further enhanced with advanced features such as AI recommendation chatbot support, email notifications, and analytics dashboards. In conclusion, the Job Portal Management System achieves its primary goal of creating an efficient, user-centric marketplace for employment. It stands as a practical and to sure platform that is well-aligned with the demands of the modern digital economy. With its modular architecture, the system is prepared for future scalability and the integration of emerging technologies, ensuring its long-term .

6.2 Future Enhancements

The Job Portal Management System can be further improved by adding advanced features [11] enhance user experience and system will efficiency. Future enhance ments may include suitable jobs based on skills and experience. Chatbot integration can be added [14] provide instant support and answer user queries. Email and SMS notifications can be implemented for job alerts, interview schedules, and application status updates.

Video interview scheduling and online assessment mod ules can also be intro duced to the simplify hiring process. Advanced analytics dashboards [11] employers and administrators can help track these recruitment performance and user activity. Mobile application support for Android and iOS can increase accessibility, while stronger security features such as OTP verification and two-factor authentication can improve data protection.

In addition [15] functional upgrades, future versions of the system will focus on optimizing backend performance through microservices architecture, By adopting the continuous integration [13] continuous deployment (CI/CD) pipeline, the plat form will be able to roll out these updates seamlessly, availability while evolving to meet the dynamic demands of the modern digital recruitment landscape.

From a technical perspective, the system's architecture will be transitioned to ward a Microservices model to improve fault tolerance and independent scalability of modules. This transition will be supported by a CI/CD (Continuous Integration / the Continuous Deployment) pipeline, allowing for seamless updates and zero down time deployments. Finally, to ensure maximum accessibility, a dedicated mobile application for Android and iOS platforms will be developed using Flutter or React Native, ensuring that job seekers can stay connected and apply for opportunities while on the go.

Chapter 7

ADDRESSING COMPLEX ENGINEERING PROBLEM AND SDG

Mapping of the Project to Complex Engineering Problem (CEP)

Level	**Attribute**	**Characteristics**	**WKs**	**Justification**
WP1	Depth of Knowledge	Requires in-depth engineering knowledge	WK3, WK4, WK5, WK6	Requires knowledge of Spring Boot, Spring MVC, Spring Security, JPA, and database design for building a multi-role job portal system.
WP2	Conflicting Requirements	Technical and non-technical conflicts	WK4, WK5, WK6	Balances role-based access control with usability, ensuring employers and job seekers have separate secure and intuitive workflows.
WP3	Depth of Analysis	Analytical and design thinking	WK3, WK4, WK5	Involves entity-relationship modeling, REST API design, and integration of file upload and email notification services.

Level	**Attribute**	**Characteristics**	**WKs**	**Justification**
WP4	Familiarity	Partially new problems	WK4, WK8	Job-matching and application tracking introduce modern challenges in multi-role web systems beyond traditional CRUD applications.
WP5	Applicable Codes	Limited standard solutions	WK6	Requires custom design for resume storage, application status tracking, and dynamic job filtering not fully covered by standard frameworks.
WP6	Stakeholder Needs	Multiple stakeholders involved	WK5, WK6, WK7	Handles distinct requirements of job seekers, employers, and admins, ensuring secure and efficient role-specific interactions.
WP7	Interdependence	System-level integration	WK3, WK5, WK6	Integrates frontend (Thymeleaf), backend (Spring Boot), database (JPA/MySQL), file storage, security, and email notification into one unified system.

Table 7.1: Mapping of Project to Complex Engineering Problem (CEP)

Table 7.1 presents the mapping of the project to Complex Engineering Problem (CEP) by highlighting key attributes, characteristics, and their relevance in the development of the system. It explains how the project applies engineering knowledge, design thinking, and system integration to solve real-world problems. The table also shows the consideration of multiple stakeholders, security, and modern technologies in building an efficient solution.

Mapping of the Project to Complex Engineering Activities (CEA)

EA No.	Attribute	Characteristics	Justification based on the Project
EA1	Range of Resources	Use of diverse technologies	The project uses Spring Boot, Spring MVC, Spring Data JPA, MySQL/H2, Thymeleaf, Spring Security, and JavaMailSender for full-stack portal development.
EA2	Level of Interactions	Complex interactions between modules	The system manages layered interactions between job listings, applications, user authentication, resume uploads, and email notifications.
EA3	Innovation	Application of modern technologies	Integration of role-based dashboards, dynamic job filtering with JavaScript/ES6, file upload handling via MultipartFile, and email status notifications demonstrates innovation.
EA4	Consequences to Society and Environment	Impact on users and society	Improves employment accessibility, reduces manual hiring overhead, and provides a transparent and secure digital platform for job seekers and employers.
EA5	Familiarity	Beyond standard practices	Combines multi-role authentication, resume management, application tracking, and real-time dashboard analytics into a unified portal beyond traditional job boards.

Table 7.2: Mapping of Project to Complex Engineering Activities (CEA)

Table 7.2 presents the mapping of the project to Complex Engineering Activities (CEA) by outlining key attributes and their practical implementation. It highlights the use of multiple technologies, system interactions, and innovative approaches in developing the application.

SDG Mapping (CEP Section)

SDG No.	SDG Title	Relevance to the Project
SDG 8	Decent Work and Economic Growth	Facilitates employment by connecting job seekers with employers, supporting fair recruitment processes and improving career opportunities.
SDG 9	Industry, Innovation and Infrastructure	Promotes digital transformation in recruitment using Spring Boot REST APIs, dynamic filtering, and automated notification systems.
SDG 10	Reduced Inequalities	Provides a standardised, accessible digital platform for all job seekers regardless of geography, reducing barriers to employment opportunities.
SDG 16	Peace, Justice and Strong Institutions	Ensures data security and role-based access control using Spring Security and password hashing, maintaining integrity and fairness in the hiring process.

Table 7.3: SDG Mapping for the Project

Table 7.3 presents the mapping of the project to Sustainable Development Goals (SDGs) by showing its relevance to education, innovation, and sustainability. It highlights how the system improves learning opportunities, supports digital transformation, and reduces manual processes. The table also emphasizes the project's contribution to secure, efficient, and eco-friendly event management.

Chapter 8

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